Vision-Based Drone Detection in Complex Environments:
A Survey
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The researchers (Liu et al., 2024) conducted a comprehensive survey and comparative analysis of vision-based drone detection methods in complex environments.

The researchers systematically reviewed the field by some steps. First ,they Characterized Imaging and Drones by analyzing the unique mechanisms of visible and infrared imaging and identified the specific traits of drones (e.g., small size, varying types, and motion blur) that make them difficult to detect. Second they organized the main obstacles into four areas: data acquisition, backgrounds and drones, imaging environments, and algorithm efficiency. They then summarized existing solutions for each, such as multi-modal fusion for weather issues and super-resolution for small targets. Then they compiled and detailed numerous RGB and infrared (IR) drone datasets, providing a central reference for researchers to find appropriate training data.

To provide a fair comparison of existing technologies, the researchers conducted experiments evaluating representative methods in the field. They selected two widely adopted infrared small target datasets: SIRST and NUDT-SIRST. They tested a variety of algorithm types such as Filter-based: Top-Hat and Max-Median. Traditional Spatial Context-based: WSLCM and TLLCM. Sparse Low-Rank Tensor-based: IPI and RIPT. Deep Learning (Data-driven): They tested several models, including MDvsFA-cGAN (the first GAN-based method), ACM, ALCNet, UIU-Net, DNA-Net, and RepISD-Net.

The researchers used five key metrics to measure performance. The Accuracy and Error Metrics like Intersection over Union (IoU) , Probability of Detection (Pd) ,On the SIRST dataset, UIU-Net achieved the highest IoU (78.25), while both UIU-Net and RepISD-Net reached a perfect Pd of 100. On the NUDT-SIRST dataset, RepISD-Net achieved the best IoU (89.44), and UIU-Net showed the highest Pd (98.94), the False Alarm rate (Fa),Traditional methods like Top-Hat and WSLCM struggled heavily with background noise, yielding extremely high false alarm rates ,DNA-Net achieved the lowest false alarm rates across both datasets (2.51 on SIRST and 3.71 on NUDT-SIRST )

Efficiency and Complexity Metricsl ike model Parameters (M),RDIAN (0.22 M) and RepISD-Net (0.28 M) were identified as the most lightweight models in terms of learnable parameters ,MDvsFA-cGAN (60.30 M), was significantly larger ,. and FLOPs (G) ,RDIAN was the most efficient at 3.72 G, followed by DNA-Net at 14.26 G .

The researchers' testing revealed critical performance differences between the methodologies. Traditional methods (like Top-Hat or WSLCM) frequently struggled with noise in complex backgrounds, leading to significantly higher false alarm rates (Fa). Deep learning algorithms such as UIU-Net, DNA-Net, and RepISD-Net demonstrated the highest accuracy and the lowest false alarm rates across both datasets. While deep learning models were more accurate, the researchers noted that there is still significant room for optimization regarding algorithm complexity and operational efficiency for real-time hardware deployment.